



MIDDLESEX Community College

Tools and Technologies for Tech Writers 2025

Week 11

Sample Bookmap

Middlesex Community College 2025

This document was prepared as an assignment for the Middlesex Community College Tools and Technologies for Technical Writers class, Winter semester 2025.

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The name is defined by the value of the author key defined in your bookmap. I set them up to be your name.

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Building a PC by Robert Palmer

A basic guide to buying components and assembling a personal computer

Introduction

Why build a desktop PC?

The primary reason to build a desktop is to get a system tailored to your needs as a user. While some prebuilt systems can offer good value for the cost, they still won't match your needs as well as a properly researched home-built computer will. Building a desktop has become easier and more affordable over the years, with many modern builds requiring little more than a Phillips screwdriver and your hands to assemble.

However, while anyone can build their own desktop, not everyone needs to do so. If all you need is basic web browsing and an office suite then a laptop or pre-built desktop would cover your needs for less money. Moreover, a prebuilt laptop provides the mobility that your built desktop cannot. And if your needs are simple, the cost of a basic laptop might be comparable to building a desktop.

In general, if you need more power and flexibility than a basic office PC and don't mind being mostly tethered to a desk, then building a desktop PC is the right choice.

How Hard is it?

How difficult is it to build a desktop PC?

Building a desktop is not as complex as it might seem at first. Most parts and cables snap together with a little pressure or only require a few screws to hold in place. Connectors are generally clearly labeled and designed to only go together one way, making it difficult to go wrong during assembly. It's not substantially more difficult than assembling a LEGO set or building an IKEA bookshelf, although the specter of financial risk when placing a \$200 processor into a \$100 motherboard can make it feel like more of an ordeal.

The most difficult part of building a desktop is not the labor required to put everything together, but the planning that goes into selecting parts. The number of possible components and configurations can very easily seem overwhelming. The reader needs the ability to gauge their own needs, set a budget, do basic internet research, handle simple tools, and follow basic directions.

Planning the Build

Determine your budget and purpose before going shopping

The number of options in computer parts available is at the same time one of the prime reasons to build your own desktop and one of the prime obstacles to overcome. It is very easy to become overwhelmed with options without a plan, and just buying whatever looks good or is on sale is a sure route to compatibility issues and spending money on performance the rest of your machine cannot use. You need a plan before you start shopping for parts. Consider the following points before diving into parts list:

1. What is the desktop for?

What are you going to do with your desktop that requires more power than a laptop or pre-built desktop? Do you need a powerful graphics card for 3D modelling and cutting-edge gaming graphics, or can you get by with a lesser option?

2. What is your budget?

Costs for building a desktop can vary wildly, from a few hundred dollars to several thousand at the high end. Setting an overall target budget before you start will inform the general performance range to look at and roughly how much to spend on each component. There's no point buying a high end \$900 graphics card for a \$1,000 total budget build, you won't even be able to afford a power supply and case!

3. Can you reuse any parts?

If you already have a desktop, some of the parts may be reusable, saving money out of your budget that can then be applied to other parts. Broadly speaking, all peripherals, such as a keyboard, mouse, monitor, or speakers, can be reused without problem assuming they still fit your needs. Legacy equipment, like a PS/2 mouse, might need replacement or at least an adapter for modern systems, but generally it's not a problem. Some internal components can also be reused, but you will have to do research to see if they are compatible. In general, do not reuse power supplies, because they have limited lives and power supply faults can damage your new parts.

4. Avoid bottlenecks

Computers are limited by their slowest part, something commonly called **bottlenecking**. A \$900 graphics card is wasted on a build with a \$50 processor and \$50 worth of RAM because the rest of the machine will not be able to keep up with it. It is better to approximately match performance across parts so no one component is holding your desktop back.

5. Futureproofing

I recommend planning your build around currently available parts that meet your current needs. In general, it's not worth trying to anticipate future hardware trends and overspend on components now in order to stay ahead of the curve later. Cutting edge technology usually comes at a premium and will be outdated in six months. Pricing and availability are also influenced as much by geopolitics as advances in technology, meaning it can be difficult to guess how much a component will cost a year from now or what the state of the art will be.

Gratuitous Table

to meet the requirements

I couldn't think of anywhere to put a table, so its here

Look at this table

Its got text	So much text
I'm not really a 'table person'	<i>I'll put some random formatting too</i>
I like plain text	Here's an odd one

Components

Exploring what components go into a desktop PC

A desktop PC consists of several parts. The bare minimum for a working computer is a central processing unit, motherboard, random access memory, power supply unit, and a monitor and keyboard for interaction. In

practice though you will want a case to put everything in, a storage device, and an operating system. Graphics and networking cards will round out your machine's capabilities.

Components are presented in approximate order of importance to your build. Your choice of CPU will determine what motherboard you need, and those two components are the core of your build. The next most important part is the graphics card, followed by the RAM. Networking cards, storage, and the case are vital but are rarely major choice points. The PSU is presented last of the major components not because it is least, but because you'll need to know your total wattage requirements from every other part before selecting it.



Figure 1: Typical desktop components, Credit: J.P. Rickabaugh

Shopping for Components

How to determine what component models and specifications will your desktop build use

Now that you have a strong idea of what you want out of your build project it's time to start looking at what components are available. There are several sites that help group together parts into general performance tiers, such as [Logical Increments](#), but it's important to take any rating as a suggestion and do your own follow up research, especially if you're looking at a larger build budget. It's worth repeating that computer components can be a volatile industry, and new technology, geopolitical shifts, and consumer trends can have an enormous impact in a short period of time.

One of the most popular resources is [PCPartPicker](#), a site that compiles information on computer part pricing and reviews and has robust planning tools. The most useful tool is the Builder, which allows you to select parts from a sortable list and add them to your shopping list. The site does not sell parts directly, so adding a part to your list will generate an affiliate link to the current least expensive supplier. As you add items to your list future items can be filtered by compatibility using an option on the left side of the screen (on by default). This ensures parts are at least advertised to work together, although physical dimensions aren't checked. As you fill out your shopping list, the site automatically generates a link you can save or share with others.

Computer building is a popular hobby and there are several online forums, including PCPartPicker's own forum, who gladly will discuss your choices in detail and at great length. More in-depth reviews of parts can be found online. Some reputable sites for reviews are [AnandTech](#), [TechPowerUp](#), and [TechSpot](#).

In general, I would recommend never buying a used PSU, and being very cautious about buying a used GPU. GPUs are often used in bulk for continuous repetitive tasks, like cryptography, and then swapped out and sold used when new technology becomes cost effective. A used GPU, even a very recent one, may have been working 24 hours a day, 7 days a week for months before being resold. It may have a deceptive amount of wear and tear with a significantly shortened life. Other components are fine to buy used if you can find a good discount from a reputable seller.

Where to buy components

Components can be purchased either online or from physical retailers

Once you've finalized your parts list finding a good price from a seller with stock can be a challenging task. Prices can fluctuate over a matter of months depending on new technology, political circumstances, and changes in demand. Two parts that are near identical may vary significantly in price due to branding or seller.

The majority of sellers in the US operate through online storefronts rather than keeping stock in physical stores. While you might find a small local parts seller near you, there is only one brick and mortar retailer of note in the US: Micro Center.

Online

The largest online computer parts suppliers in the US are Amazon and Newegg with Best Buy being a distant third, but their offerings and policies have become very similar over time. Newegg has a slightly more generous return policy with no restocking fees for returned goods, while Amazon inconsistently charges a percentage. In general, though, go with whichever merchant is listed on PCPartPicker's parts list at the time to take advantage of sales.

Physical

[Micro Center](#) is the only nationwide brick and mortar computer parts retailer left in the US, but it has held on for a reason. Micro Center frequently has bundles and deals that nothing online can compare to, but such deals are almost always in store pick up only. Willingness and ability to travel to a Micro Center can result in significantly better parts for less than what would be available online or at any other retailer. Micro Center currently has 28 stores spread across the US in 19 states.

Putting it Together

Planning a build, acquiring parts, and assembling a desktop

This will provide a worked example of determining what kind of desktop PC we want, finding the parts for the build, and then assembling the desktop.

1. [Plan the build](#)
2. [Shop for components](#)
3. Assemble the parts
4. Enjoy your PC

Example Planning

A worked example of a build plan

The first step in building a desktop is to develop a plan

1. What is the desktop for?
I want to build a new mid-range PC that can also play games at good settings for the next few years.
2. What is your budget?
I can spend about \$1,000 USD
3. Can you reuse any parts?
My current PC is very old and most of its parts are too obsolete for re-use. I can salvage my case, storage drive, monitor, and other peripherals though.

Example Shopping

A worked example of shopping for components

The second step in building a desktop is to find and purchase your build components

1. I go to [Logical Increments](#) and look for a build set that roughly totals out to \$1,000. This is the ‘Great’ category. I won’t use these suggestions exactly, but they are a good place to start looking.



RX 6750XT	\$300	RS 7600	\$182	Pure Rock 2	\$38	ASRock B650 PG	\$150	8GB DDR5	\$23	2TB	\$65	512GB	\$48	SeaSonic 650W (B)	\$70	4000D	\$90	\$966
RX 6700XT	\$310	RS 7600X	\$195	Peerless	\$40	MSI B650-P	\$168	16GB DDR5	\$47	4TB	\$85	1TB	\$70	EVGA 650W (G)	\$87	Meshify C	\$100	
RTX 3060 Ti	\$500			NH-U125 Rdx	\$55	GIGABYTE B650 Aorus	\$200			8TB	\$165	2TB	\$97	EVGA 650W (G)	\$254	P500A	\$110	
expand to see details																		

2. I decide to buy online rather than go to Micro Center.
3. I then go to [PCPartPicker](#) and click on Start Your Build in the middle of the page.

- I go through each category and pick parts roughly in line with my budget. I know the GPU will be the most expensive part of the build, followed by the CPU and motherboard. I want all my parts to be roughly in the same performance tier, so for a \$250 GPU I want an approximately \$200 CPU and motherboard.

Component	Selection	Base	Promo	Shipping	Tax	Price	Where	
CPU	AMD Ryzen 5 7600X 4.7 GHz 6-Core Processor	\$194.00	—	✓ Prime	—	\$194.00	amazon.com	Buy ×
CPU Cooler	Thermalright Peerless Assassin 120 SE 66.17 CFM CPU Cooler	\$34.90	—	✓ Prime	—	\$34.90	amazon.com	Buy ×
Motherboard	MSI B650 GAMING PLUS WIFI ATX AM5 Motherboard	\$169.99	—	FREE	—	\$169.99	BH	Buy ×
Memory	G.Skill Flare X5 32 GB (2 x 16 GB) DDR5-6000 CL30 Memory	\$99.99	—	✓ Prime	—	\$99.99	amazon.com	Buy ×
	+ Add Additional Memory							
Storage	Crucial P3 Plus 1 TB M.2-2280 PCIe 4.0 X4 NVME Solid State Drive	\$59.99	—	✓ Prime	—	\$59.99	amazon.com	Buy ×
	+ Add Additional Storage							
Video Card	Sapphire PULSE Radeon RX 7600 8 GB Video Card	\$269.99	—	FREE	—	\$269.99	newegg	Buy ×
	+ Add Another Video Card							
Case	+ Choose A Case							
Power Supply	Gigabyte UD750GM 750 W 80+ Gold Certified Fully Modular ATX Power Supply	\$79.98	—	✓ Prime	—	\$79.98	amazon.com	Buy ×

- I decide on getting an additional SSD to go with my old HDD so my new system isn't bottlenecked by old slow storage. I will install my OS and any games on the SSD while all my music stays on my HDD.
- The peak power draw for the system is estimated at 409W. I selected a 750W PSU for better efficiency.
- My total comes to \$908.84. I could either be satisfied or see if I can find a few more dollars to get a new monitor for my new graphics card.

Assembly

Putting all the parts together

Modern PCs are very easy to put together. Cards slot into their sockets only one way. Power cables also only connect one way. The main hurdles are getting the CPU and heatsink installed, which usually requires applying thermal paste, screwing the base together, and then snapping the fans onto the heat sink in a cramped space, and then arranging the cables such that you can see what you're doing when inserting RAM and other cards into your motherboard.

Prepare the Area

Make assembly easier and safer with a prepared area

- Make sure your work area is clean, well lit, and has a non-conductive surface to work on. This will make finding dropped screws much easier, and prevent accidental electrostatic discharge from damaging your components while assembling. An electrostatic discharge wristband, which grounds your body so you don't break sensitive electronics with a static shock, is nice to have but not strictly necessary as most components have a thin coating to mitigate static discharge. It's under \$10 on Amazon if you're really worried though. You can also invest into an electrostatic dissipative mat and rubber soled shoes if you're very concerned.

2. Gather your tools

- Thermal Paste
- Phillip's head screwdriver
- Zip or twist ties
- Tweezers
- Flashlight
- Electrostatic Discharge Wristband (optional)
- Bootable media for installing your OS

Assembly

It is finally time to start connecting components

You should never force any components together. All connectors are keyed to only go together one way, and will eventually snap together with moderate pressure. Getting them apart again usually involves lifting a latch or tab and can be quite difficult.

Turn off the PSU and unplug the cable from the socket before adjusting any cabling or components. Never install or remove any component while the PSU is connected to the wall outlet.

1. First install the CPU into the socket on the motherboard according to the CPU instructions.
 - a) Do not force the CPU into the socket. If it doesn't settle in with light pressure check the alignment of the CPU with socket and try again.
 - b) If it still doesn't settle in re-verify you have the right type of motherboard socket for your CPU.
2. Build the clips for the cooler around the CPU socket.
 - a) If you have an M.2 SSD storage device now is the best time to install it into the motherboard. This kind of device connects directly to the motherboard, and as more components are added it will be harder to get your hands in there.
3. Install the motherboard into the case.
 - a) Carefully align the holes in the motherboard with the holes in the case.
 - b) Place screws in an X pattern, starting from the corners and working inwards. After placing one screw place the next on the opposite corner. Do not tighten until all screws are placed.
 - c) Gently tighten screws in the same X pattern, again starting from the corners and working inwards.
 - d) Do not overtighten the screws. This can damage the motherboard.
4. Instal the PSU, but do NOT plug it in yet.
5. Carefully install the GPU, RAM, and any other cards into their respective slots.
 - a) Do not use excessive force. All components should slot into place with steady medium pressure.
 - b) Ensure the clips that hold the RAM cards in place are snug. Improperly slotted RAM can damage the RAM and cause the system to not boot up.
6. Install fans into the case if they did not already come screwed in.
 - a. Intake fans should be on the front of the case
 - b. Exhaust fans should be on the rear of the case

- c. The difference between an intake fan and an exhaust fan is usually just orientation. There is an arrow on the fan to indicate airflow direction.
- 7. Connect case fans, lights, and any other peripherals to the motherboard.
- 8. Connect the PSU to the motherboard, GPU, and any other peripherals.
 - a) Double check the PSU is turned off and unplugged before connecting or disconnecting any cables.
 - b) All power cables should snap into place with firm, medium pressure.
- 9. Finish installing the heatsink and cooler according to their directions.
 - a) You only need a pea sized amount of thermal paste for the heatsink.
 - b) I save this step for last because the cooler will block access to a large section of the motherboard and make connecting any cables much more difficult.
- 10. Tidy up the cabling.
 - a) Zip ties or twist ties work to keep cables secured.
 - b) In particular make sure cable are clear of any fans.

Turning it On For the First Time

Flipping the switch for the first time is the moment of truth for your build

You've planned and built your computer, and now it's time to flip the switch! This is easily the most stressful and exciting part of the whole process, when you turn the power supply on, hit the power button on the case, and hope for the best.

If all has gone well the screen will come alive with the Power On Self-Test (POST) screen.

If instead, the computer just beeps at you, take note of the beeps. Those are an error message that can be used to diagnose the problem. Some motherboards have an LED display for error codes that can be helpful in troubleshooting.

If nothing happens the problem is most likely with the power supply: either the PSU itself is bad or produces insufficient wattage, the cabling is not fully connected, or you have forgotten to turn it on.

Anything short of shooting sparks is probably solvable eventually. In the worst-case scenario you've learned a lot, and Amazon, Newegg, and Microcenter will accept returns within 30 days.

If everything works you can close the case up, usually by sliding a panel into place and then tightening thumb screws on the rear. Congratulations!

Installing the OS

The final step is to install an OS to make your desktop usable

Follow installation instructions to install your operating system to finish up your desktop PC build. You will need either a bootable DVD or a bootable flash drive with the OS installation files and an internet connection.

Glossary

Processor and Cooler

The central processing unit (CPU) is the brain of the computer. Its power will affect every program you run, and is generally the most important part of your computer and the second most expensive. The two

major manufacturers of consumer processors are Intel and AMD. Both are roughly equivalent but are not interchangeable. Intel chips only fit into Intel sockets, and AMD chips only fit into AMD sockets.

Overclocking is a method of tuning your processor to get more performance from it, but it's purely optional. Intel chips ending with K are overclockable and usually come at a cost premium, while almost all AMD chips can be overclocked.

Processors can get very hot while running and need to be cooled to avoid damage. They usually come with small heat sinks, which are arrays of metal fins installed over the processor to help dissipate heat. However, the heatsinks that come with processors are usually inadequate for heavy usage. You should buy a better cooling solution. Installing this is likely the single most complicated part of building a new computer. Most cooler kits come with thermal paste for installation, but if yours doesn't you will need to buy a small tube as well.

Motherboard

The motherboard, or mainboard, is the main circuit board inside your computer which connects together all of the other parts. There are many manufacturers, with the most popular being ASUS, ASRock, GIGABYTE, and MSI.

Your motherboard's processor socket type must match the CPU and fit inside your case. An Intel chip will not fit in a motherboard designed for AMD chips, and you must also match the socket type. For instance, an Intel i5-12500 processor needs an LGA 1700 socket on the motherboard.

Your board will also need enough slots for everything you want to install, although that is rarely a concern. Also, take note of the form factor of your motherboard, such as ATX or mini-ITX. This determines the limits on size and power for the motherboard. For a first-time builder I would recommend a micro-ATX or ATX form factor for their flexibility.

Graphics Processing Unit

The graphics processing unit is a dedicated processor, separate from the CPU, that draws images and graphics. The power of your GPU will determine how good displayed images look. This is likely the most or second most expensive part in your computer if you want to play games or do any kind of video editing or modeling.

GPUs are good at doing large numbers of repetitive tasks, such as drawing hundreds of image frames per second with only slight differences. This also makes GPUs decent at cryptography and generative AI application, although not as good as a specially designed dedicated component. But because GPUs are good enough at those tasks they can be subject to high price volatility whenever a new technology takes off. More than any other component price and availability for GPUs can vary wildly.

There are two major GPU competitors, Nvidia and AMD. As a general rule Nvidia pushes performance with a price tag to match while AMD offers more budget conscious GPUs.

Random Access Memory

If the CPU is your system's brain, then the RAM is how many thoughts your system can hold on to at one time. A very fast CPU will need a large amount of RAM to meet its full potential.

RAM is categorized by the basic technology and clock speed, e.g. DDR4-3200. Higher numbers are better for both cases; DDR5 RAM is better than DDR4, and a 3200 clock speed is better than 2133. Most RAM can be overclocked for better performance. Make sure your RAM is compatible with your motherboard.

Networking

You will most likely want to connect your new desktop PC to the internet or a home network. Some motherboards have onboard ethernet and wi-fi, but not all of them do. Thinking ahead and buying a \$20 networking card can save a few frustrating days waiting for one to show up in the mail after ordering it on your phone.

Storage

You will need somewhere to put your files. The most common options are spinning disk Hard Disk Drives (HDDs) and Solid-State Drives (SSDs). HDDs tend to be less expensive while SSDs are significantly faster. One option would be to get two drives, a small but fast SSD for your operating system and a larger HDD for bulk storage for things like movies and photos.

Case

You don't technically need a case; however, it will make things much safer and more organized to have everything tucked away inside a case. Cases also allow you to direct airflow using fans in order to keep your components cool.

The most important consideration when getting a case is the form factor. Your case needs to be at least large enough to fit your motherboard and other components. An ATX form factor case is large enough for an ATX motherboard, but can also mount a smaller micro-ATX motherboard. Larger cases also have more room to work in while installing components.

The second most important consideration is how it looks. This is what you will be looking at every time you use the computer. There are cases with lights, transparent panels, and all kinds of color schemes. Or you can get a basic beige box.

Ventilation is also a concern, but most cases will have enough mounting brackets for fans. If a case does not come with fans, buy enough to have at least one intake fan and one exhaust fan. More fans can provide better cooling, albeit with diminishing returns, but they will also create more noise.

Power Supply Unit

The Power Supply Unit converts wall outlet power into power for your machine. It connects to your motherboard and has additional connectors for other components such as hard drives and GPUs. A modular PSU means that cables can be removed as needed, making cable management much easier and airflow better at a small premium.

How much wattage you will need for your PSU depends on the components in your system. You can add up the total power draw from each component, which will be listed in each component's technical manual, and then add extra wattage to help with efficiency. Power supplies are typically most efficient at around 50% of nominal capacity, so if your total power draw is 300 watts then a 600-watt PSU will be most efficient. A PSU constantly operating near maximum load will generate a lot of heat and require noisy fans to cool, and will also wear out faster and put your system at risk.

It is very important to get a trusted, high-quality PSU. Defects in your PSU can lead to your whole system being destroyed by power surges. Do not reuse a PSU, and try to remind yourself to replace your PSU every five years or so, especially on high end systems. An "80 Plus" rating can be an indicator of quality and efficiency. When buying a PSU don't rely on just the brand name, look at the overall reviews for each model.

Peripherals

At a minimum you will need a mouse, keyboard, and monitor. Those do not come free with the case outside of a Micro Center bundle. You might also consider getting a DVD drive to make installing software on your new machine easier. When looking at monitors keep in mind how much you're investing into your GPU. If you're spending \$900 on a high-end GPU you will also want a high-end monitor to display your crisp graphics.

Operating System

The operating system (OS) runs the computer and allows you to run programs on your hardware. The most popular operating system is Windows, although alternatives such as Linux exist.

Technically, Original Equipment Manufacturer (OEM) licenses are only available to companies that build computers, not individual hobbyists. However, it is easy to get an OEM Windows license for half the price of a retail license from reputable retailers such as Wal-Mart.

Whether you choose Windows or another option you will need some method of installing your OS on your new machine. You can either use a DVD drive or an 8 GB+ flash drive with a disc image installed on it.